

Software Requirements Specification

Francanglais Compiler and Dataset Platform

Requirements, User Stories and Agile Delivery Plan

Group members and matricules:

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Group submission. Complete the names and matricules before submission.

Planning basis: four proposed two-week sprints, selected by the project owner. Story points, task hours and suggested leads are estimates, not records of completed work.

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1 Introduction and Document Control

1.1 Purpose and audience

This SRS defines the expected behavior, quality constraints and delivery backlogs of the Francanglais (Camfranglais) platform. It supports contributors, the project group, reviewers and testers. It accompanies the Software Design Description (SDD), which explains the architecture and UML workflows.

1.2 Evidence and requirement status

The baseline is grounded in `data_collector/App.py`, `data_collector/dataset.py` and `data_collector/audio_utils.py`. The root README describes later language-analysis ambitions. The coursework JSON contains an explicitly illustrative grammar, not a validated corpus grammar.

Status	Meaning throughout this document
B	Baseline behavior is supported by available source. This is not a claim that every acceptance test has been executed.
T	Target requirement or proposed enhancement; implementation or release-level verification is still required.
D	Deferred enhancement, retained in the product backlog but not committed to the four-sprint proposal.

Compiled frontend assets and Python caches do not establish a supported backend API or a complete compiler implementation. No HTTP route, classifier accuracy, completed sprint or measured team velocity is inferred from them. This revision adds acceptance criteria, user stories, a prioritized product backlog and four sprint backlogs to the initial specification.

1.3 Definitions

Entry

A word, phrase or sentence with annotation fields and optional audio.

Gloss

A contributor-supplied meaning in standard French or English.

Backlog item

A prioritized unit of product work, identified by PB.

User story

An actor goal, its benefit and acceptance criteria, identified by US.

Story point (SP)

A relative complexity estimate; it is not an hour or a measured velocity.

Sprint backlog

Selected product items and the concrete tasks proposed for one two-week iteration.

2 Overall Description

2.1 Product scope

The implemented product is a single-workstation desktop collector with Collect, Browse & Edit, and Stats tabs. It stores UTF-8 CSV data and local audio files. It has no authentication or concurrent-writer coordination. Contributor and maintainer are usage roles, not enforced permission levels.

The proposed next increment consumes corpus evidence to tokenize input, classify tokens and explain sentence-level analysis. It must remain separate from data-entry operations. A future predictor for unseen words is deferred. A web service, automatic transcription, production linguistic accuracy and multi-user database deployment are outside the four-sprint commitment.

2.2 Stakeholders and environment

- **Contributor:** captures linguistic examples and optional recordings.
- **Maintainer:** corrects annotations, removes unwanted rows and checks corpus coverage.
- **Learner/analysis user:** requests the proposed language-analysis results.
- **Project group/reviewer:** verifies acceptance evidence and documentation.

Python, CustomTkinter and Tcl/Tk provide the desktop UI. NumPy, sounddevice, soundfile and a working PortAudio/device environment support recording. Attaching an existing file does not require a working microphone. The current workstation is Windows; platform playback branches are not proof of tested support on every operating system.

3 Functional Requirements

The acceptance checks below are the required verification outcomes. Baseline rows describe existing behavior accurately; target rows must not be read as already implemented.

ID Status	Requirement and acceptance check
FR-01 <i>B</i>	The system shall initialize the managed audio directory and create the CSV header if the dataset is absent. Acceptance: Start with missing storage: both are created. Start with existing data: rows and header remain unchanged.
FR-02 <i>B</i>	Saving shall reject text that is empty after surrounding whitespace is removed. Acceptance: Submit spaces or blank text: show “Missing text” and append no row. Non-empty text can proceed.

ID Status	Requirement and acceptance check
FR-03 <i>B</i>	The collector shall offer word, phrase and sentence entry types and the ten defined categories. Acceptance: Each option is selectable and the chosen value is stored; defaults are word and greeting after a form reset.
FR-04 <i>B</i>	The collector shall accept optional French gloss, English gloss, notes and contributor text. Category and type use selection controls. Acceptance: Save with blank optional fields and again with annotations; values round-trip without making glosses mandatory.
FR-05 <i>B</i>	A new row shall receive a generated ten-character hexadecimal UUID-derived ID and a local ISO-8601 timestamp with seconds. Acceptance: Verify format on insertion and that editing preserves ID and timestamp. Collision prevention is not implemented.
FR-06 <i>B</i>	Saving shall append one UTF-8 CSV row using the ten-field schema. Updating or deleting shall rewrite the canonical header and retained rows. Acceptance: Read the file with a CSV parser after each operation; field order and unrelated rows are preserved.
FR-07 <i>B</i>	The Collect tab shall show an advisory duplicate warning for equal text after strip/lower normalization, scheduled after 400 ms of typing inactivity. Acceptance: A case/outer-space variant warns; a distinct value clears the warning. A duplicate can still be saved; it is not a uniqueness rule.
FR-08 <i>B</i>	When recording support and a microphone work, Record/Stop shall capture mono audio at 44,100 Hz and stage a WAV filename. Acceptance: Capture frames, stop and verify the WAV. With no frames, show “No audio captured” and create no new recording.
FR-09 <i>B</i>	The contributor shall be able to choose an existing audio file or cancel the chooser without changing the draft. Acceptance: Select a file to stage its copy; cancel and verify the previous pending filename remains unchanged.
FR-10 <i>B</i>	New recordings and attachments shall use generated filenames inside the managed audio directory; attachments preserve the original extension. Acceptance: Verify the copied/recorded file and the saved row’s filename. The original attached file is not modified.

ID Status	Requirement and acceptance check
FR-11 <i>B</i>	Browse shall filter rows using a case-insensitive substring of text, category or contributor and populate the editor from a selected cached row. Acceptance: An empty query lists all rows, a non-match lists none, and selection displays the corresponding metadata without a second read in the selection handler.
FR-12 <i>B</i>	A selected entry shall support editing text, type, category, both glosses and notes, or deletion from the CSV. Acceptance: An edit changes only the allowed fields; deletion removes matching ID rows. The ID, contributor, timestamp and audio filename are not editable here.
FR-13 <i>B</i>	Pending or selected audio shall be playable through the audio libraries, with an OS-player fallback when unavailable or unsuccessful. Acceptance: A missing file returns failure; a failed library attempt can reach the OS fallback; failure of both paths produces the GUI error dialog.
FR-14 <i>B</i>	An explicit Stats refresh shall show total rows and descending counts/bars by entry type and category. Acceptance: Compare with a known fixture; empty data shows zero total and “No data yet.” No divide-by-zero occurs.
FR-15 <i>T</i>	The proposed analysis increment shall accept text and return tokens, token labels with evidence, and a sentence-analysis status with diagnostics. Acceptance: Approved mixed-language fixtures yield reproducible structured results; empty/unsupported input has explicit diagnostics rather than fabricated success.
FR-16 <i>T</i>	Analysis shall be read-only with respect to the source dataset and audio. Acceptance: File hashes before and after successful and rejected analysis requests are identical. New predictions are not automatically inserted.
FR-17 <i>B</i>	After a successful save, the collector shall refresh count/recent entries and clear the draft while retaining contributor. Escape also retains contributor; Clear may remove it. Acceptance: Verify the latest six rows appear newest-first, form defaults reset, pending audio clears, and the contributor follows the invoked reset action.

ID Status	Requirement and acceptance check
FR-18 <i>B</i>	Edit/delete without a selection shall show an information message; deletion of a selection shall require Yes/No confirmation. Acceptance: No selection or No leaves the CSV unchanged; Yes removes the row and clears the selected ID/audio reference. The audio file itself remains.
FR-19 <i>T</i>	A failed CSV mutation shall preserve the last valid dataset and report an actionable error instead of showing saved/deleted success. Acceptance: Inject write/replace failures: the prior file remains parseable and byte-identical, and the UI reports the failure. Current direct rewrites do not satisfy this target.
FR-20 <i>D</i>	A future unknown-word predictor shall distinguish speculative candidates from corpus-confirmed entries and provide its evidence or abstain. Acceptance: Unseen-token fixtures never appear as confirmed corpus facts; prediction does not mutate the dataset. No accuracy claim is made without evaluation.

4 Non-Functional Requirements

Proposed thresholds are acceptance targets, not measured results.

ID Status	Quality requirement and verification
NFR-01 <i>B</i>	The desktop shall enforce a minimum window size of 760 by 620 pixels and offer Ctrl+Enter for Save and Escape for clearing the draft. Acceptance: Resize and exercise all tabs/shortcuts; record any clipped essential controls as a release defect.
NFR-02 <i>B</i>	Storage shall use UTF-8 and proper CSV quoting rather than delimiter concatenation. Acceptance: Round-trip accented text, embedded commas, quotes and line breaks with no field loss.
NFR-03 <i>B</i>	Playback results shall distinguish failure from accepted playback/launch requests. Acceptance: Mock missing files and library/OS exceptions; failure returns false and displays an error. True means request accepted, not sound heard to completion.

ID Status	Quality requirement and verification
NFR-04 <i>B</i>	Managed paths shall be based on the collector directory and generated media filenames, not persistent links to user-selected source files. Acceptance: Change the process working directory and attach a file; storage still resolves locally and the source media remains intact.
NFR-05 <i>B</i>	Failure to import optional audio dependencies shall disable live recording without preventing text collection, browsing or file attachment. Acceptance: Run a dependency-unavailable fixture and verify those non-recording operations. Device-open failures are a separate robustness concern.
NFR-06 <i>T</i>	Analysis shall be deterministic for a fixed corpus, grammar and rule configuration. Acceptance: Repeat each golden fixture ten times and compare serialized tokens, labels and diagnostics, excluding intentionally variable request metadata.
NFR-07 <i>T</i>	Critical collector paths shall have automated regression tests using isolated temporary data and mocked audio hardware. Acceptance: All approved CRUD, validation, duplicate, fallback and failure cases pass without modifying the team's real corpus. No test suite completion is assumed.
NFR-08 <i>T</i>	Submission artifacts shall exclude credentials and raw personal recordings unless explicitly approved for release. Collection and sharing shall follow documented consent. Acceptance: Review exported files and repository exclusions; retain consent/approval evidence separately with restricted access. Ignoring a file does not remove it from old history.
NFR-09 <i>B</i>	Documentation shall build locally from LaTeX and PlantUML sources with reproducible diagram references and readable PDFs. Acceptance: Render every UML source, compile both documents, and check images, references, table layout and placeholders using the documented commands.
NFR-10 <i>T</i>	On a recorded reference Windows workstation with 8 GB RAM and local SSD, search and Stats refresh shall each have p95 latency at most 1 second for 1,000 rows of up to 500 text characters. Acceptance: Measure 20 warm runs per operation after one warm-up, excluding audio capture; report hardware/runtime and the nearest-rank 95th percentile. This target is not benchmarked here.

ID Status	Quality requirement and verification
NFR-11 <i>T</i>	CSV updates shall use failure-safe replacement, and recording/device failures shall produce useful feedback without a false success status. Acceptance: Fault-inject disk and audio-device failures; preserve valid data, permit retry, and record the visible diagnostic. Baseline writes and recording handlers have gaps.
NFR-12 <i>T</i>	The team shall maintain a manual backup/restore procedure for the CSV and referenced audio; backup copies shall have restricted access. Acceptance: Restore a fixture backup to a separate directory and compare all rows and referenced-file hashes. No backup button is claimed.

5 Data and External Interface Requirements

Field	Meaning and constraint
<code>id</code>	Generated ten-hex-character row identifier; no collision check is currently enforced.
<code>text</code>	Required on GUI save/edit; surrounding whitespace is removed.
<code>entry_type</code>	One of word, phrase, sentence in the GUI.
<code>french_gloss,</code> <code>english_gloss</code>	Optional contributor translations, not automatic translation output.
<code>category</code>	One selected category; unknown externally edited values are not centrally validated by the CSV helpers.
<code>notes</code>	Optional context, region or usage annotation.
<code>audio_filename</code>	Empty or one managed filename; file existence is not guaranteed by the CSV schema.
<code>contributor</code>	Optional contributor text; not an authenticated account or group matricule.
<code>timestamp</code>	Local ISO-8601 timestamp without an explicit time-zone offset; unchanged by edit.

Categories are greeting, everyday, slang, insult/banter, market/money, school/campus, food, family, proverb/expression and other. The GUI offers audio-file filters, but attachment is a copy operation, not a validated codec conversion. Local filesystem, microphone and OS-player interfaces are supported boundaries. The future analysis contract has no implemented endpoint in this snapshot.

6 User Stories and Acceptance Criteria

The stories describe desired user value. B stories still need the proposed regression evidence; T and D stories require additional development. Each story maps to exactly one product backlog item with the same numeric suffix.

6.1 US-01 — Initialize a usable corpus

As a contributor, I want storage initialized without overwriting existing entries so that I can safely start or resume collection.

Trace: FR-01, FR-06, NFR-02, NFR-04.

Acceptance criteria

- a) Given absent storage, launch creates the canonical header and audio directory.
- b) Given existing UTF-8 rows, relaunch preserves them byte-for-byte until an explicit mutation.

6.2 US-02 — Capture an annotated example

As a contributor, I want to save a word, phrase or sentence with optional meanings and context so that the corpus retains useful linguistic evidence.

Trace: FR-02–FR-06, FR-17, NFR-01.

Acceptance criteria

- a) Blank/whitespace text is rejected without a new row.
- b) Valid text with either blank or completed optional annotations saves with generated ID/time and chosen type/category.
- c) A successful save refreshes recent/count displays and resets the draft while retaining contributor.

6.3 US-03 — Notice duplicate text

As a contributor, I want a duplicate warning so that I can review repetitions without losing the ability to record legitimate variants.

Trace: FR-07.

Acceptance criteria

- a) A strip/lower-equivalent existing text warns after the typing debounce.
- b) Changing to distinct text clears the warning; ignoring it does not block saving.

6.4 US-04 — Find and select an entry

As a maintainer, I want to search text, category or contributor so that I can quickly locate an entry for review.

Trace: FR-11, NFR-10.

Acceptance criteria

- a) Case variants of a matching query return the same matching rows.
- b) Empty/non-matching queries show all/no rows respectively.
- c) Selecting a displayed row loads its own fields and audio availability.

6.5 US-05 — Correct an annotation

As a maintainer, I want to edit selected linguistic fields so that mistakes can be corrected without changing entry identity.

Trace: FR-06, FR-12, FR-18.

Acceptance criteria

- a) No selection or blank replacement text produces a message and no write.
- b) A valid edit survives reload; ID, timestamp, contributor and audio reference stay unchanged.

6.6 US-06 — Delete with confirmation

As a maintainer, I want confirmation before deleting an entry so that a mistaken click does not remove corpus data.

Trace: FR-12, FR-18.

Acceptance criteria

- a) No selection or a No response leaves the file unchanged.
- b) Yes removes matching ID rows, clears the selected reference and refreshes the display.
- c) The associated audio file remains available for separately managed cleanup.

6.7 US-07 — Attach existing audio

As a contributor, I want to attach a recording from disk so that I can contribute audio without a working microphone.

Trace: FR-09, FR-10, NFR-04, NFR-05.

Acceptance criteria

- a) Selection copies the source under a generated filename with its extension.
- b) Cancellation changes neither the draft attachment nor the original file.
- c) Saving the draft links the staged filename to the new row.

6.8 US-08 — Listen to an example

As a contributor, I want to play pending or stored audio so that I can check the recording linked to an entry.

Trace: FR-13, NFR-03.

Acceptance criteria

- a) Library playback is attempted when available; a failure can fall back to the OS player.
- b) A missing file or rejected fallback shows a playback error, not a success message.
- c) Playback never modifies CSV rows.

6.9 US-09 — Record an example

As a contributor, I want to record and stop microphone audio so that a pronunciation example can accompany the text.

Trace: FR-08, FR-10, NFR-05, NFR-11.

Acceptance criteria

- a) A working recorder produces a mono 44,100 Hz WAV and stages its filename.
- b) No frames produce a visible empty-capture message and no new WAV.
- c) Missing dependencies disable recording; device-failure feedback is target hardening, not baseline behavior.

6.10 US-10 — Review corpus coverage

As a maintainer, I want total/type/category statistics so that I can identify underrepresented categories during collection.

Trace: FR-14, NFR-10.

Acceptance criteria

- a) A refresh matches independently calculated fixture totals and groups.
- b) Empty data shows zero/no-data labels, and bars never divide by zero.

6.11 US-11 — Protect corpus quality and privacy

As a maintainer, I want regression checks, safe writes and a recovery procedure so that the team's data remains usable and appropriately protected.

Trace: FR-19, NFR-07, NFR-08, NFR-10–NFR-12.

Acceptance criteria

- a) Injected CSV write failures preserve the last valid file and report an error.
- b) Regression/performance reports use fixtures, not personal recordings.
- c) A documented backup restores with matching rows and audio hashes; publication has approval.

6.12 US-12 — Build understandable documentation

As a group member, I want editable UML and LaTeX documents with explanations so that the design and requirements can be reviewed and reproduced.

Trace: NFR-09.

Acceptance criteria

- a) Every diagram has a source, rendered image, explanation and trace to a feature.
- b) Both PDFs compile with no unresolved references or overflowing tables.
- c) The group can fill its names/matricules without changing system requirements.

6.13 US-13 — Inspect lexical analysis

As a learner, I want tokenization and evidence-based labels so that I can see how mixed-language input relates to the corpus.

Trace: FR-15, FR-16, NFR-06.

Acceptance criteria

- a) Approved fixtures produce tokens with labels and evidence; unknowns remain explicit.
- b) Repeated analysis is deterministic and leaves corpus/audio hashes unchanged.

6.14 US-14 — Inspect sentence analysis

As a learner, I want sentence status and explanatory diagnostics so that I can understand the limits of the declared analysis rules.

Trace: FR-15, FR-16, NFR-06.

Acceptance criteria

- a) Valid and rejected teaching examples return an explicit status and reason.
- b) The selected grammar/rule version is identified; the starter grammar is not presented as a validated description of all Camfranglais.
- c) Empty and unsupported input is handled explicitly without modifying evidence.

6.15 US-15 — Receive a demonstrable release

As a reviewer, I want a repeatable end-to-end demonstration and acceptance report so that I can distinguish verified outcomes from unfinished features.

Trace: FR-15, FR-16, FR-19, NFR-07–NFR-12.

Acceptance criteria

- a) The selected collector and analysis stories pass their approved acceptance checks.
- b) The handover includes built PDFs, test/performance evidence, recovery steps and an honest known-limitations list.
- c) Every completed claim points to evidence; unfinished work stays in the backlog.

6.16 US-16 — Explore unseen-word candidates

As a learner, I want speculative candidate explanations for unknown words so that I can explore possible blends without mistaking a guess for a corpus fact.

Trace: FR-20, FR-16, NFR-06.

Acceptance criteria

- a) Predictions are labelled speculative with evidence or abstention.
- b) No automatic insertion or unevaluated accuracy claim is made.
- c) This story remains deferred beyond the four proposed sprints.

7 Prioritized Product Backlog

7.1 Prioritization and estimation policy

Items are ordered for delivery and dependency resolution. Must denotes a release-critical item, Should a valuable planned item, and Could a deferred option. The point estimates are relative and provisional. B means source exists to review/test, not that the item is sprint-Done. Each PB item inherits its US acceptance criteria and the referenced requirements.

ID	Story	Priority	SP	Sprint	Depends on	Work item and baseline state
PB-01	US-01	Must	3	S1	None	Storage/schema initialization (B); verify non-destructive startup.
PB-02	US-02	Must	5	S1	PB-01	Annotated entry creation (B); verify validation, metadata and reset.
PB-03	US-03	Should	3	S1	PB-02	Advisory duplicate detection (B); verify debounce and non-blocking behavior.
PB-04	US-04	Must	5	S1	PB-01	Search and cached selection (B); add regression/performance fixtures.
PB-05	US-05	Must	5	S2	PB-04	Selected-entry editing (B); verify field preservation.
PB-06	US-06	Must	3	S2	PB-04	Confirmed deletion (B); verify cancellation and media retention.
PB-07	US-07	Should	3	S2	PB-02	Audio attachment (B); verify copying and cancellation.
PB-08	US-08	Should	5	S2	PB-07	Playback/fallback (B); verify return/error semantics.
PB-09	US-09	Should	5	S3	PB-07, PB-08	Recording (B) plus target device-error hardening (T).
PB-10	US-10	Should	3	S3	PB-01	Statistics (B); verify grouping and empty corpus.
PB-11	US-11	Must	5	S3	PB-05, PB-06, PB-08, PB-09	Safe writes, tests, privacy, backup and benchmarks (T).
PB-12	US-12	Must	3	S3	PB-11	Documentation/build (B); update evidence and known gaps.
PB-13	US-13	Must	8	S4	PB-01, PB-11	Lexical-analysis increment (T); no verified compiler source.
PB-14	US-14	Must	8	S4	PB-13	Sentence diagnostics and rule provenance (T).
PB-15	US-15	Must	3	S4	PB-12, PB-13, PB-14	Release acceptance and group handover (T).
PB-16	US-16	Could	8	Later	PB-13, PB-14	Unknown-word predictor (D); evaluation corpus required.

The four-sprint scope totals **67 SP**; the deferred item adds **8 SP**, for a product backlog total of **75 SP**. Estimates are not a promise of throughput. Split or re-estimate items after team refinement without removing their acceptance conditions.

8 Sprint Backlogs: Four Proposed Two-Week Iterations

8.1 Planning assumptions and team placeholders

The schedule spans relative weeks 1–8 with no invented calendar dates. The first three sprints review, test and harden the existing collector; Sprint 4 proposes an analysis increment. This is not a reconstruction of historical sprints.

M1, M2 and M3 refer respectively to the three blank group-member rows on the title page. Leads below are **suggested and unconfirmed**. Every task initially has status **Planned**. Task hours are independent effort estimates, not a conversion from story points. No team capacity or velocity has been measured.

Sprint	Window	Goal	SP	Hours
S1	Weeks 1–2	Reliable text collection and discovery	16	24
S2	Weeks 3–4	Controlled maintenance and existing audio	16	26
S3	Weeks 5–6	Recording, corpus quality and documentation	16	29
S4	Weeks 7–8	Read-only analysis and verified handover	19	27

8.2 Sprint 1 backlog — Collection and discovery

Selected: PB-01–PB-04 (16 SP, 24 estimated hours). Start with a schema fixture and isolated test directories.

Task	PB	Task and completion evidence	Lead	h
S1-T01	PB-01	Review initialization/schema; create missing/existing-storage fixtures.	M1	3
S1-T02	PB-01	Automate UTF-8, quoting and non-destructive startup checks.	M3	3
S1-T03	PB-02	Review form fields, non-empty validation, defaults and generated metadata.	M2	4
S1-T04	PB-02	Verify save/reload, recent rows, contributor retention and shortcuts.	M3	3
S1-T05	PB-03	Review 400 ms debounce, cancellation of earlier jobs and normalization.	M1	2
S1-T06	PB-03	Test warning appearance/clearance and permitted duplicate saves.	M3	2
S1-T07	PB-04	Review filter/cache selection and create representative search fixtures.	M2	4
S1-T08	PB-04	Test empty, mixed-case and non-match queries; capture initial timings.	M3	3

Review/exit: demonstrate a valid save, rejected blank save, advisory duplicate and matching search. All four stories have acceptance evidence; unresolved defects remain visible rather than marked Done.

8.3 Sprint 2 backlog — Maintenance and audio files

Selected: PB-05–PB-08 (16 SP, 26 estimated hours). Depends on S1 storage, selection and save behavior.

Task	PB	Task and completion evidence	Lead	h
S2-T01	PB-05	Review editable-field whitelist and read-modify-write behavior.	M1	4
S2-T02	PB-05	Test edit persistence, no selection, blank text and preserved metadata.	M3	4
S2-T03	PB-06	Review confirmation, ID filtering and selected-state cleanup.	M2	3
S2-T04	PB-06	Test Yes/No paths and unchanged audio files after row deletion.	M3	3
S2-T05	PB-07	Review chooser cancellation, UUID naming and copying without microphone.	M1	3
S2-T06	PB-07	Test original-file preservation and staged filename persistence.	M3	3
S2-T07	PB-08	Review library playback and platform launch fallback boundaries.	M2	3
S2-T08	PB-08	Mock missing media, library errors and OS launch rejection.	M3	3

Review/exit: demonstrate edit, cancelled/confirmed deletion, attachment and failed-playback feedback. Verify no unwanted media deletion.

8.4 Sprint 3 backlog — Quality, recording and documentation

Selected: PB-09–PB-12 (16 SP, 29 estimated hours). This sprint explicitly includes work not present in the baseline, especially failure-safe persistence and device-error feedback.

Task	PB	Task and completion evidence	Lead	h
S3-T01	PB-09	Review stream lifecycle and add explicit device-failure feedback.	M1	3
S3-T02	PB-09	Test frames/no frames, disabled dependencies and failed device opening.	M3	4
S3-T03	PB-10	Review totals, field grouping and normalized descending bars.	M2	3
S3-T04	PB-10	Test empty data, category fixtures and explicit Stats refresh.	M3	2
S3-T05	PB-11	Implement/test atomic CSV replacement and truthful write-failure messages.	M1	6

Task	PB	Task and completion evidence	Lead	h
S3-T06	PB-11	Run regression/performance checks; rehearse restore and review consent/exclusions.	M3	4
S3-T07	PB-12	Update UML explanations, requirement links and local document build steps.	M2	4
S3-T08	PB-12	Render all diagrams; compile/review PDFs and verify backlog traceability.	M3	3

Review/exit: show recording and statistics plus fault-injection evidence, a restored backup and reproducible documentation. Compiler work does not bypass this corpus-quality gate.

8.5 Sprint 4 backlog — Analysis increment and release

Selected: PB-13–PB-15 (19 SP, 27 estimated hours). The team must first approve a representative fixture corpus and a declared teaching grammar/rule set. If unavailable, split/replan the analysis items; do not claim language coverage without evidence.

Task	PB	Task and completion evidence	Lead	h
S4-T01	PB-13	Implement/refine a deterministic tokenizer and evidence-based token labels.	M1	6
S4-T02	PB-13	Add mixed-language golden tests, unknown handling and read-only hash checks.	M3	4
S4-T03	PB-14	Implement declared-rule sentence analysis with provenance and diagnostics.	M2	6
S4-T04	PB-14	Test accepted/rejected/empty cases and repeated deterministic output.	M3	4
S4-T05	PB-15	Run end-to-end acceptance, failure/recovery, privacy and performance review.	M1	4
S4-T06	PB-15	Assemble PDFs, evidence, limitations and the group demonstration.	M2	3

Review/exit: demonstrate all selected stories with reproducible evidence. PB-16 stays deferred. Across the four sprints, the proposed task effort totals **106 hours**; availability must be confirmed by the team.

8.6 Definition of Ready, Done and change control

Ready: actor value, requirement links, acceptance criteria, dependency availability, fixture data/consent, estimate and lead are agreed before work starts. **Done:** implementation is reviewed, applicable automated/manual tests pass, acceptance evidence is recorded, documentation is current, and no known critical data-loss or privacy issue remains in the selected scope.

Use Planned, In progress, Blocked and Done states on the team's board. Review progress daily; hold a demonstration and retrospective after each sprint. At each boundary, re-estimate remaining work and update dependencies and totals. A source-supported

feature is not retroactively a completed sprint. New scope is prioritized in the product backlog before entering a sprint.

9 Traceability to Design and Verification

Stories / items	SDD view	Required evidence
US-01–03 / PB-01–03	Collection activity and save sequence	Initialization, validation, duplicate and UTF-8 fixtures.
US-04 / PB-04	Search/select sequence	Filter results and cache-to-selection checks.
US-05 / PB-05	Edit sequence and class view	Allowed fields change; stable metadata preserved.
US-06 / PB-06	Delete sequence and state view	Confirmation branch and audio-retention checks.
US-07 / PB-07	Save/attachment sequence	Copy/cancel and filename-link checks.
US-08 / PB-08	Playback sequence	Library/OS fallback and false-result checks.
US-09 / PB-09	Recording sequence	Stream lifecycle, WAV and device-failure checks.
US-10 / PB-10	Statistics sequence	Known totals/groups and timing report.
US-11 / PB-11	Persistence and limitations sections	Safe-write injection, backup restore and privacy review.
US-12 / PB-12	All UML views and build process	Local rendering, explanations and compiled PDFs.
US-13–14 / PB-13–14	Proposed component/deployment boundary	Future lexical/sentence fixtures and read-only checks.
US-15 / PB-15	Verification and release gates	Acceptance matrix with evidence and unresolved gaps.
US-16 / PB-16	Deferred design decisions	Later evaluation plan; no current completion claim.

10 Constraints, Risks and Open Decisions

- CSV writes are currently non-atomic and uncoordinated. The baseline is single-writer only; FR-19/NFR-11 are planned safeguards, not current guarantees.
- Truncated UUID IDs are generated, not collision-checked. External CSV edits can introduce duplicate IDs, invalid categories or missing media links.
- Recording depends on working hardware. Long recordings are buffered in memory with no stated duration limit; large-audio scaling is not promised.

- Compiler fixtures, language labels and teaching rules require team review. The illustrative grammar cannot validate all language varieties.
- Estimates, dates, member identities and actual task acceptance need team confirmation. Only the four two-week sprint structure has been selected.
- Multi-user services, authentication, speech-to-text and prediction accuracy remain future scope unless the backlog is explicitly re-approved.